

Scratch — Draw Shapes with Pen

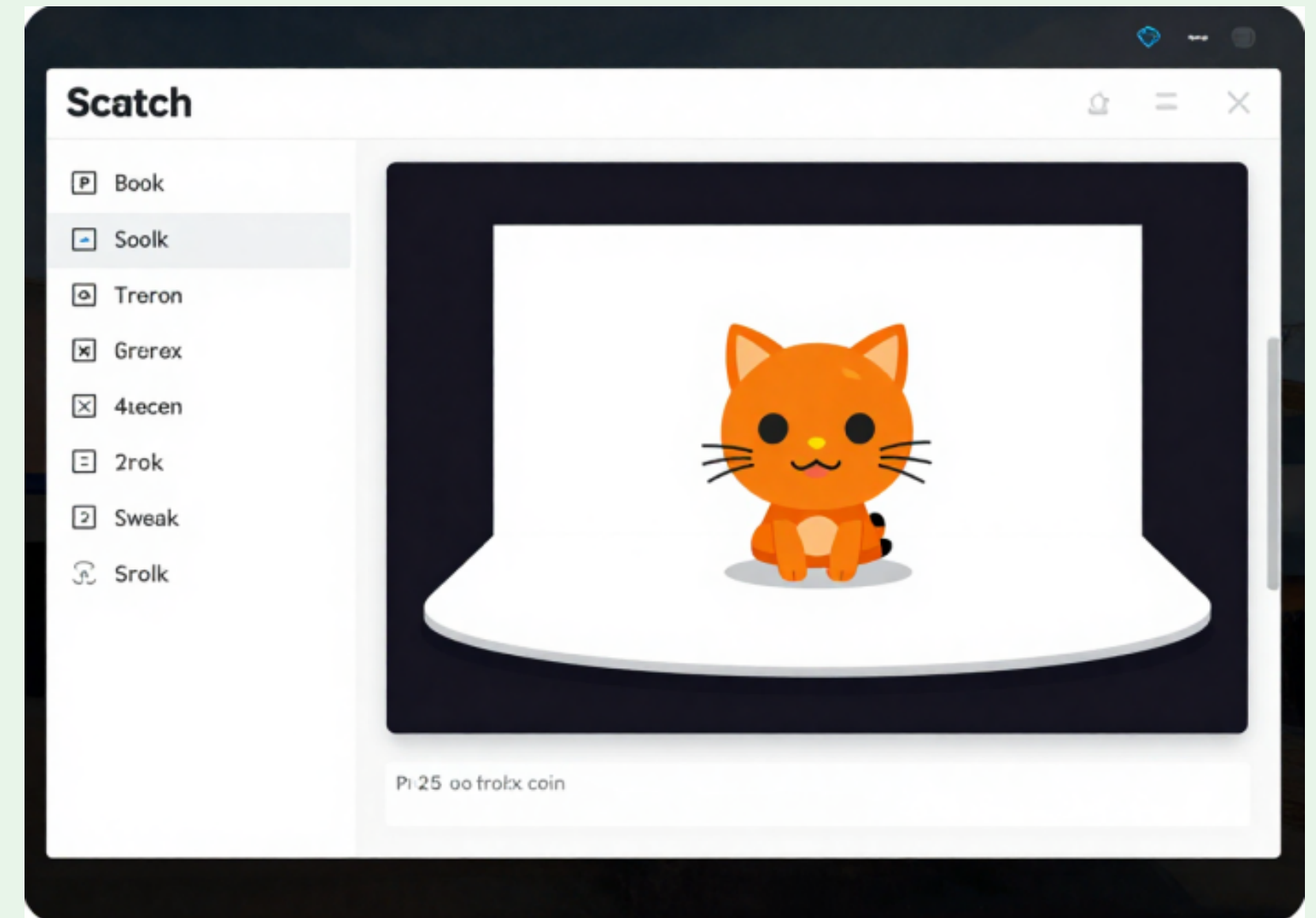
Learn to create colorful shapes using the Pen extension



What is Scratch?

Scratch is a fun way to tell stories, make games, and draw using code!

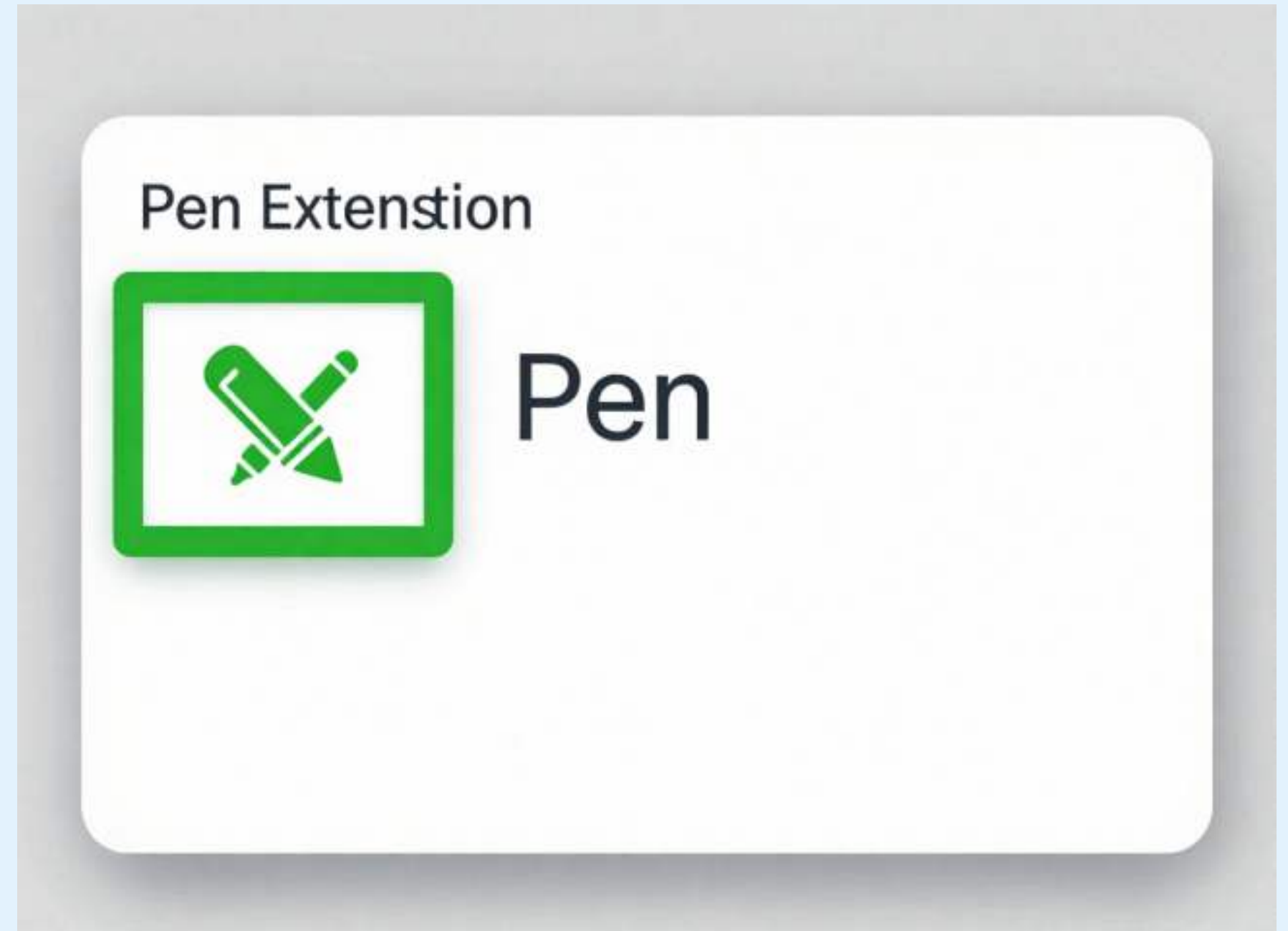
- Easy to use coding blocks
- Helps you learn how to code
- Create your own projects
- Made for kids like you!



Meet the Pen Extension

The Pen lets you draw lines and shapes on the screen!

- It works like a real pen or marker
- The sprite carries the pen as it moves
- You can change colors and thickness
- Find it in the Extensions menu

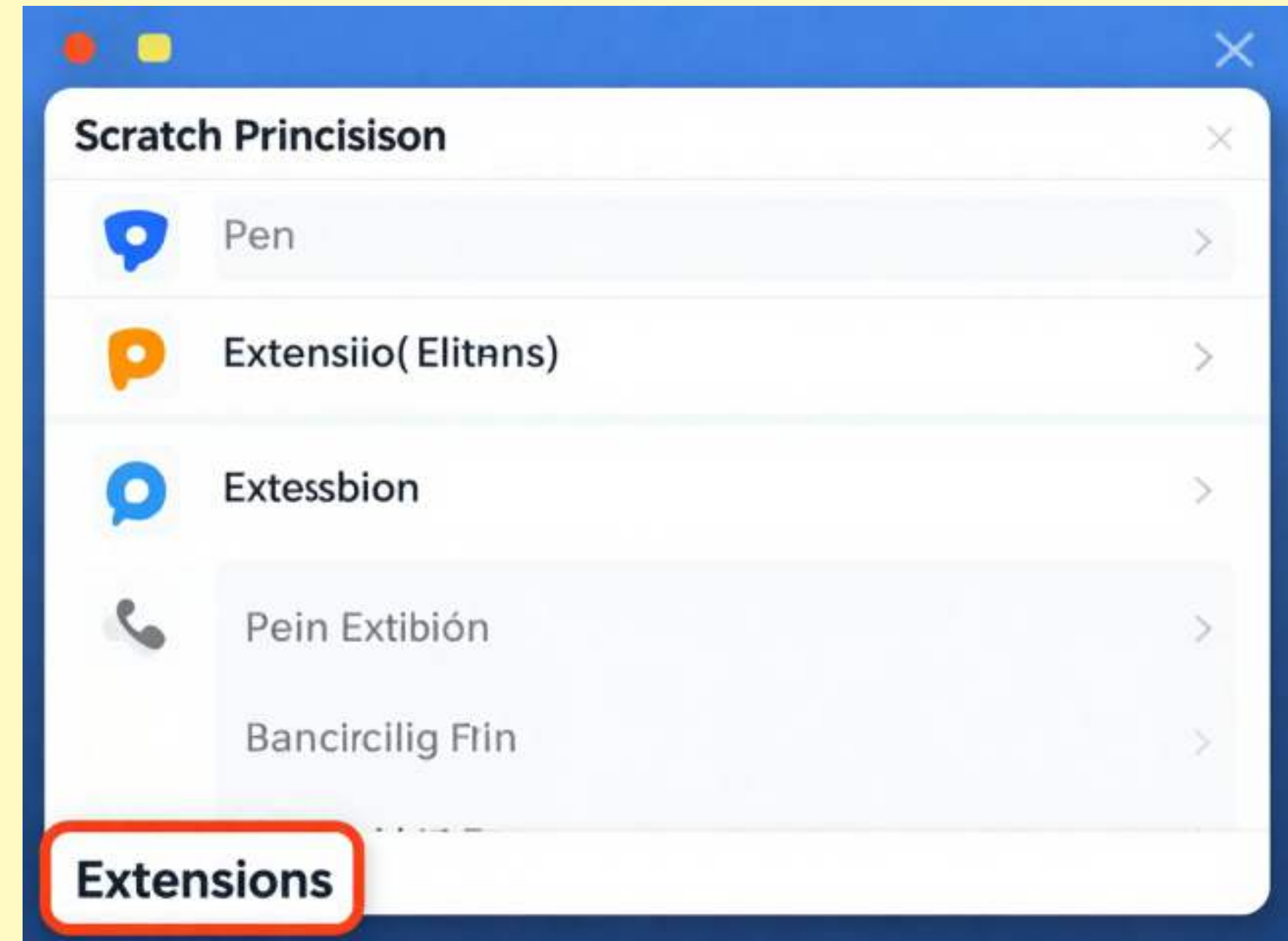


How to Add the Pen Extension

Follow these steps:

1. Open your Scratch project
2. Click the blue Extensions button (bottom-left)
3. Find and select the Pen extension
4. Pen blocks now appear in your palette!

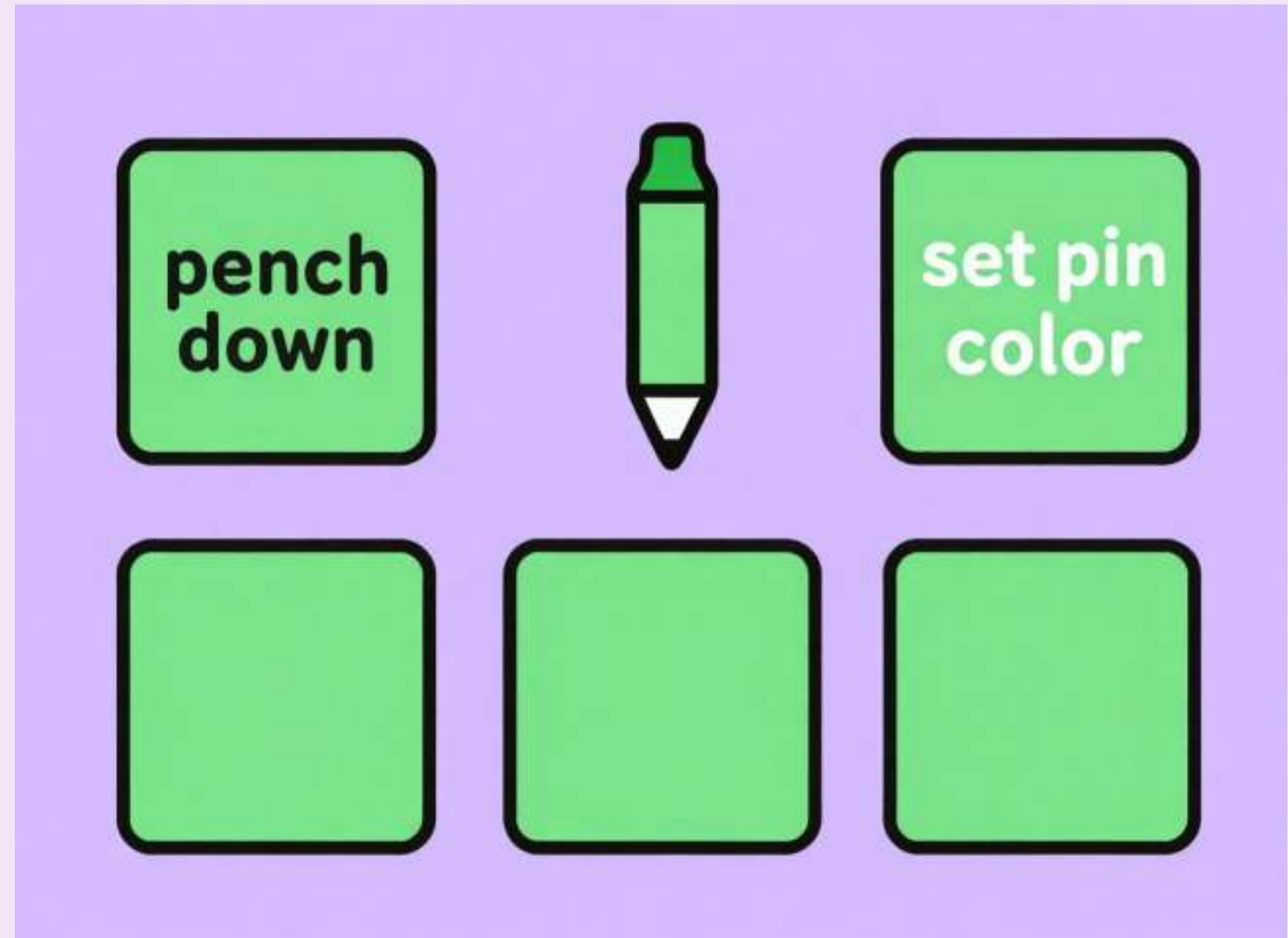
That's it — you're ready to draw!



Pen Blocks You Will Use

Here are the main Pen blocks:

- Pen Down — starts drawing
- Pen Up — stops drawing
- Set Pen Color — pick any color
- Set Pen Size — thin or thick lines
- Clear — erase all drawings
- Stamp — leave a copy of the sprite

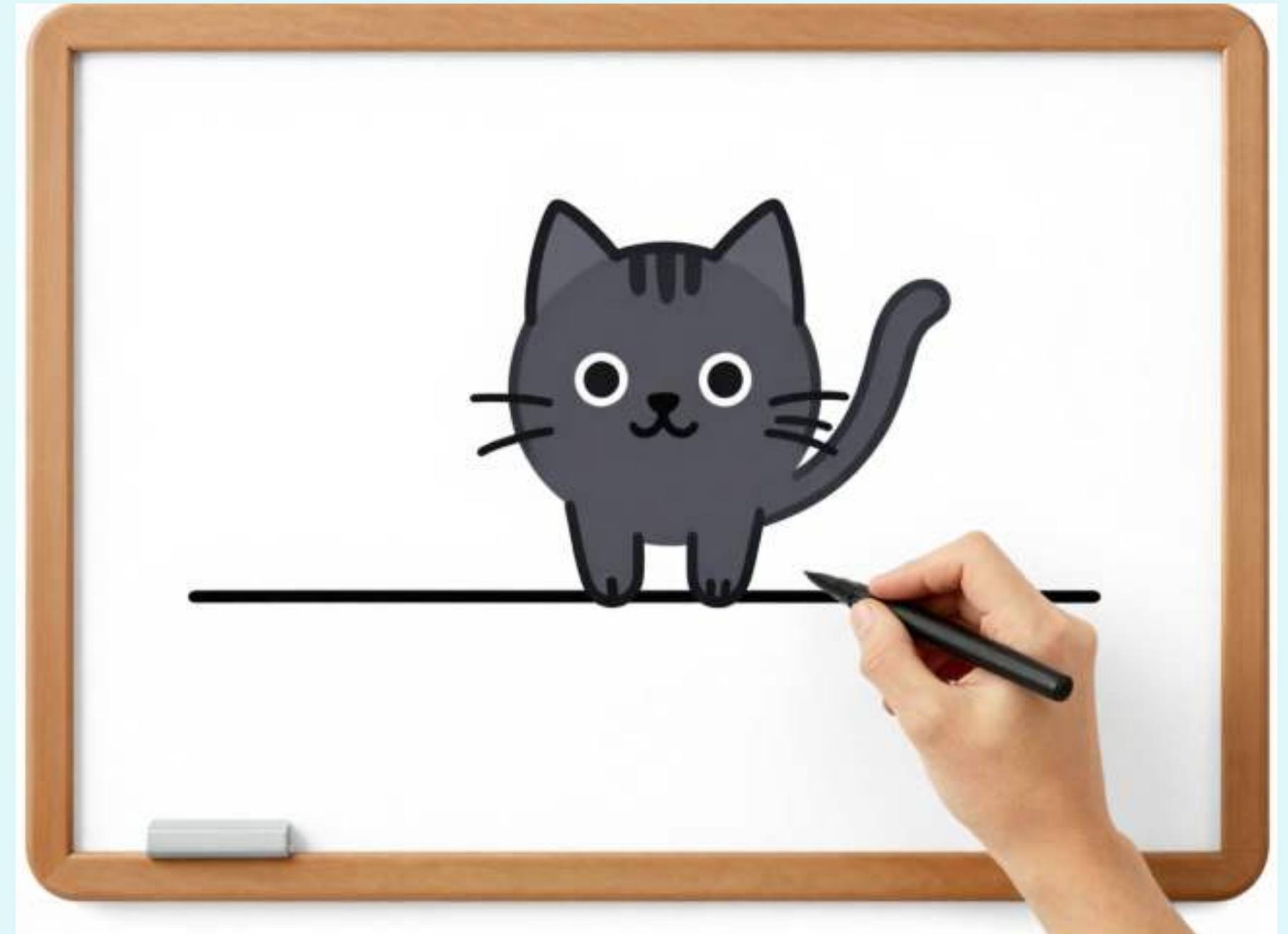


Drawing a Line with Pen Down

Try this code:

- When green flag clicked
- Clear
- Pen Down
- Move 100 steps

The sprite draws a straight line as it moves!
Pen Down means: leave a trail wherever I go.

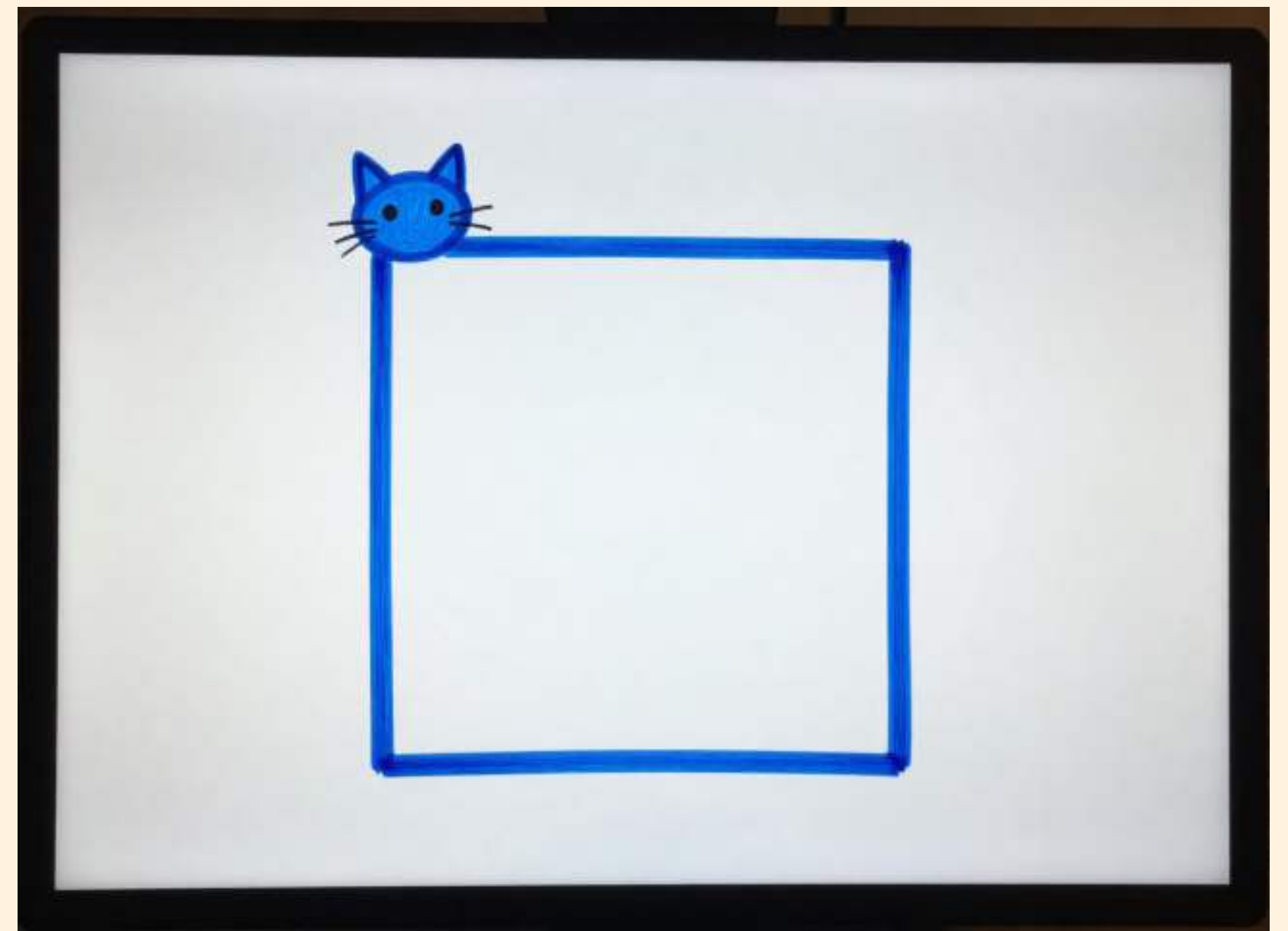


Drawing a Square

Use Repeat to draw a square:

- Pen Down
- Repeat 4 times:
 - Move 100 steps
 - Turn right 90 degrees

A square has 4 equal sides and 4 turns of 90°. The sprite draws all 4 sides automatically!



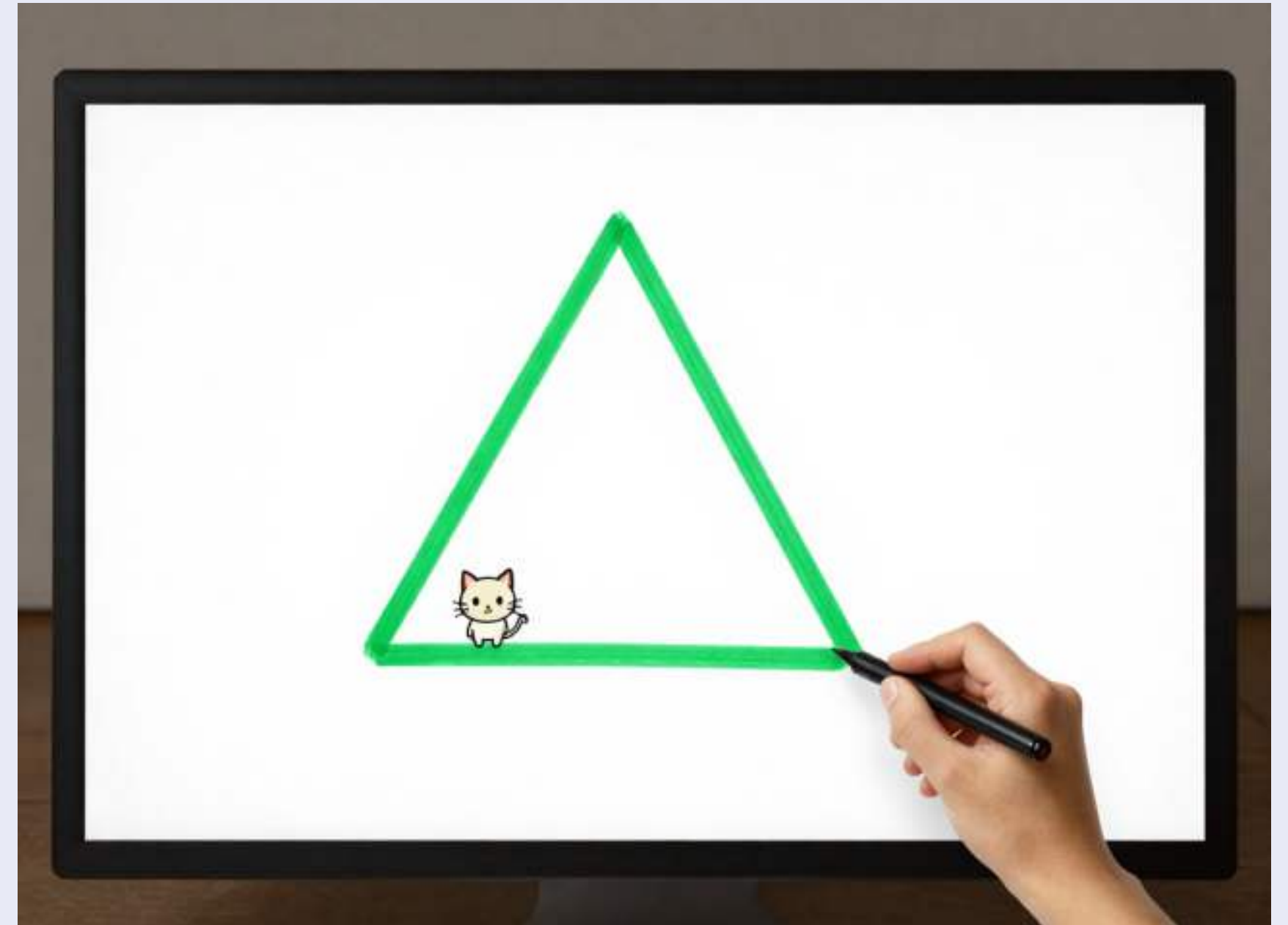
Drawing a Triangle

Draw an equilateral triangle:

- Pen Down
- Repeat 3 times:
 - Move 100 steps
 - Turn right 120 degrees

A triangle has 3 sides.

Each turn is 120° (not 60° !) because we turn the outside angle.

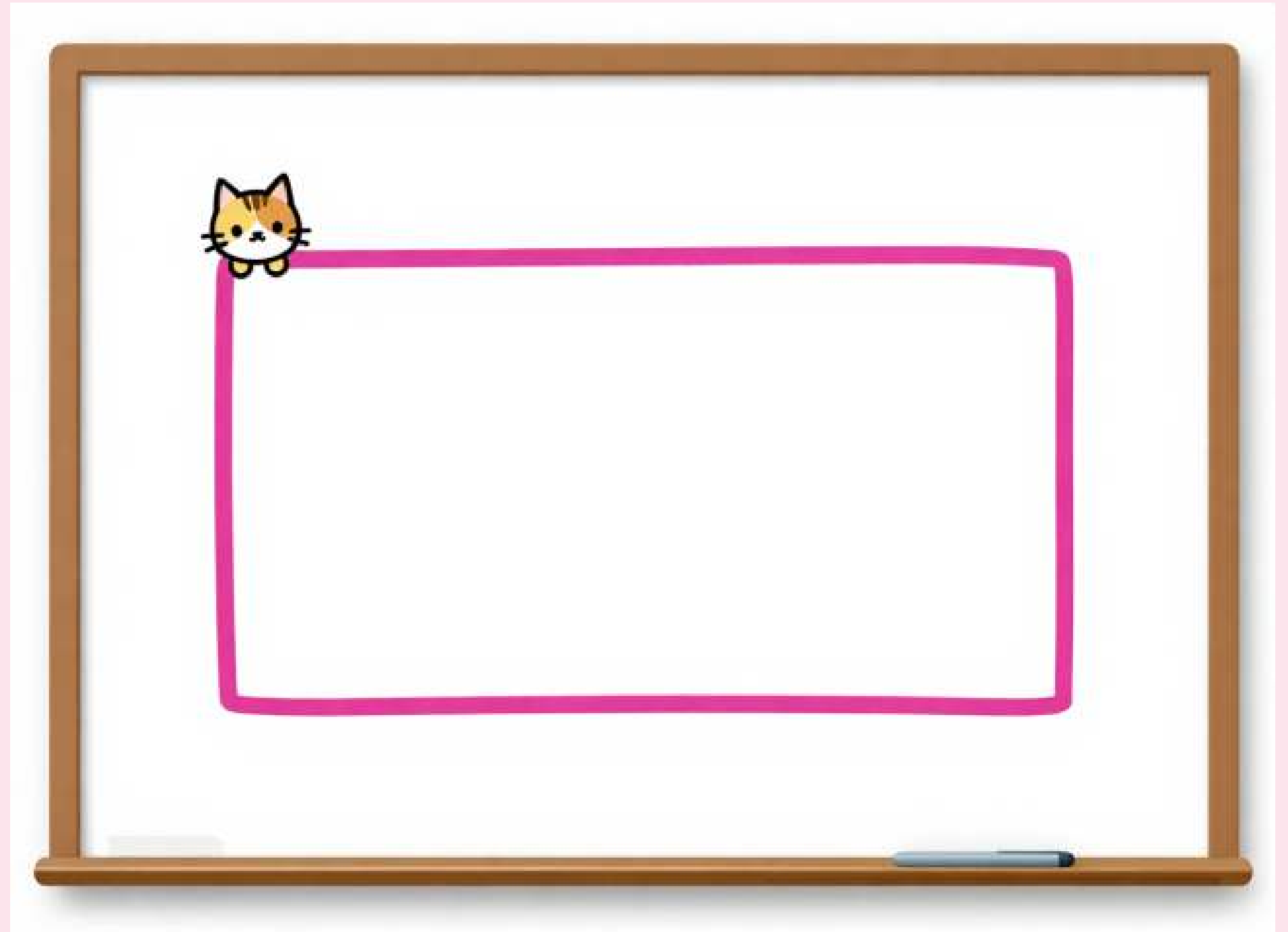


Drawing a Rectangle

Draw a rectangle with different sides:

- Pen Down
- Repeat 2 times:
 - Move 150 steps
 - Turn right 90 degrees
 - Move 80 steps
 - Turn right 90 degrees

A rectangle has 2 long sides
and 2 short sides!



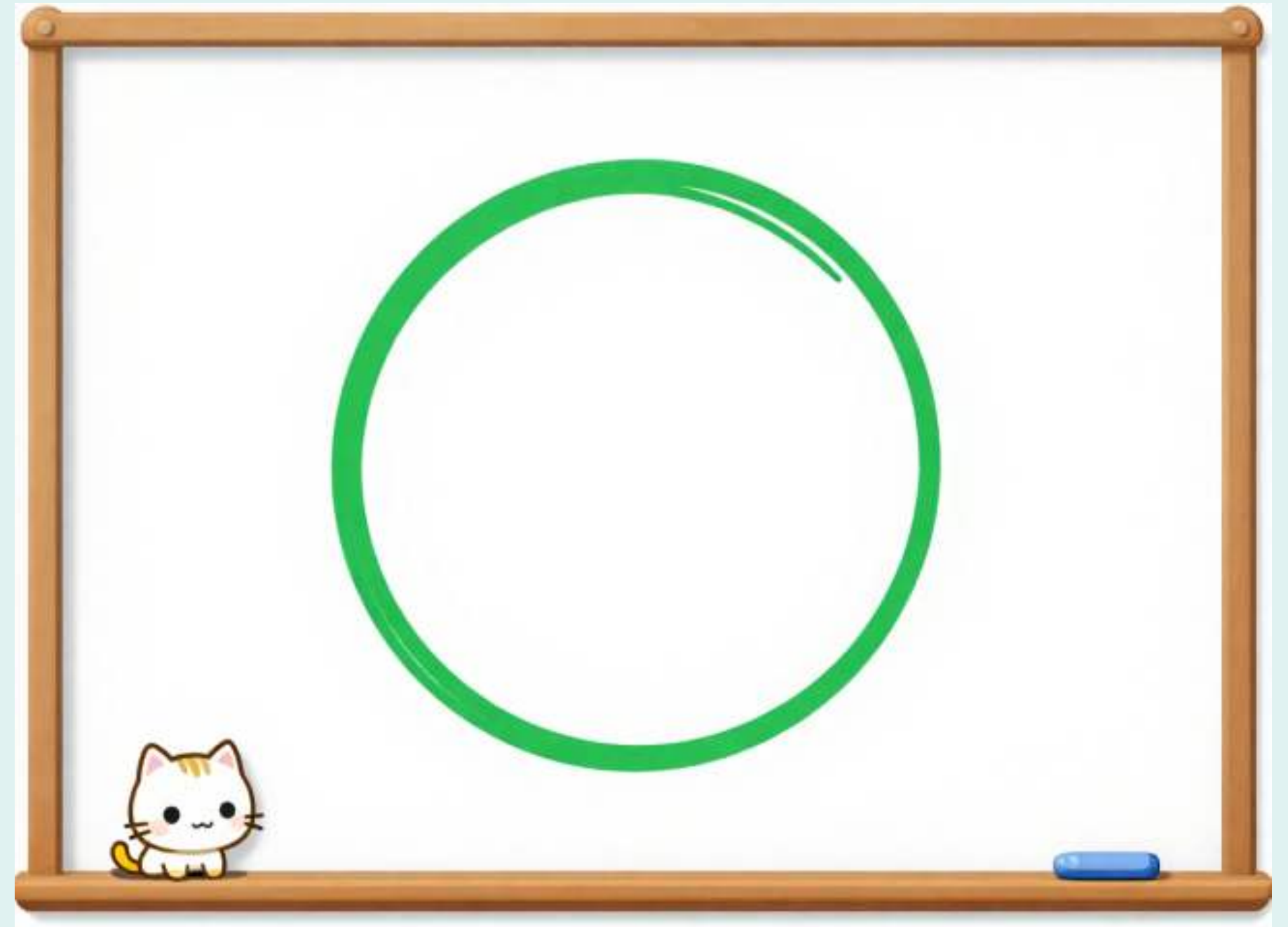
Drawing a Circle Using Steps

Trick to draw a circle:

- Pen Down
- Repeat 36 times:
 - Move 10 steps
 - Turn right 10 degrees

$$36 \times 10^\circ = 360^\circ \text{ (full circle!)}$$

Small moves + small turns = smooth curve.



Pen Up and Pen Down Commands

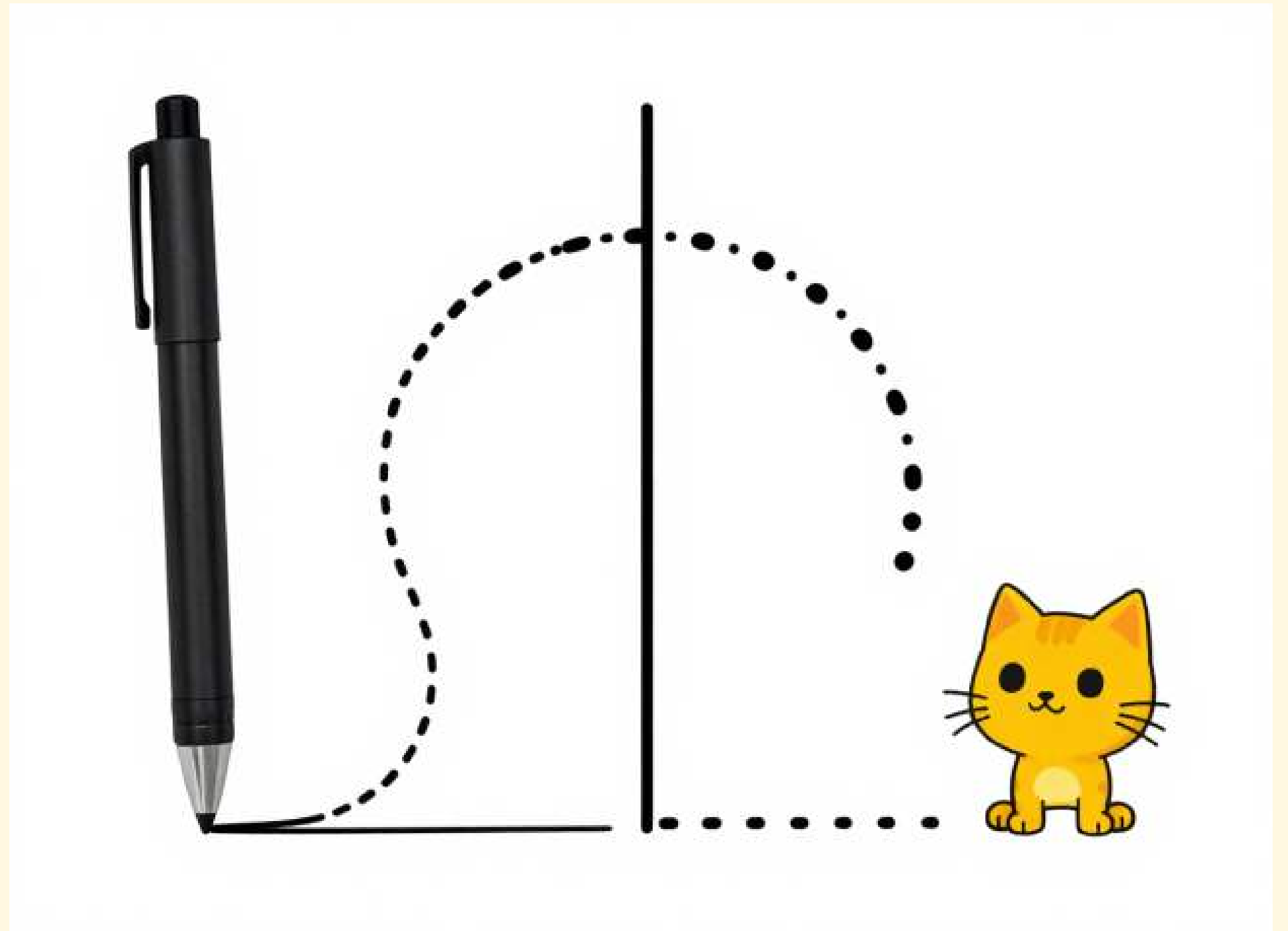
Pen Down = drawing ON

The sprite leaves a trail when it moves.

Pen Up = drawing OFF

The sprite moves without drawing.

Use Pen Up to jump to a new spot,
then Pen Down to start drawing again!



Changing Pen Color and Size

Make your drawings colorful!

- Set Pen Color — pick any color
- Change Pen Color — shift colors
- Set Pen Size — make lines thick or thin
- Change Pen Size — grow or shrink

Try rainbow lines by changing color inside a loop!



Create Fun Patterns and Designs!

Combine what you learned:

- Draw spirals by increasing move steps
- Make stars using sharp turn angles
- Create flower patterns with loops
- Mix colors for rainbow effects

Be creative — there's no wrong answer in art coding!



Hands-On Challenges

Try these fun activities:

1. Draw a house using squares and triangles
2. Make a colorful flower pattern
3. Create a star using repeat and pen blocks
4. Experiment with pen colors and sizes
5. Clear screen and start again!

Remember: Be creative and have fun!



Thank You and Keep Coding!

Great job today!

Keep exploring Scratch and drawing
amazing shapes and patterns.

Practice makes perfect!

